

Practice School II Report (Mid semester),2026

Internship Mid Term Report

PREPARED BY

Vamshi Vardhan Allapuram

2021btech010

EXTERNAL SUPERVISOR

Mr. Sajal Jain (The Correct Steps)

FACULTY SUPERVISOR

Surbhi Chhabra



NAAC 'A' Grade Accredited

Department of Computer Science & Engineering

Institute of Engineering and Technology

JK Lakshmipat University, Jaipur Rajasthan – 302026

March,2026

CERTIFICATE

This is to certify that Vamshi Vardhan (2021BTECH010), a student of B.Tech (Computer Science & Engineering), Institute of Engineering and Technology JK Lakshmi pat University,Jaipur,has satisfactorily completed the Practice School II (Mid-Semester) internship at The Correct Steps under the supervision of the undersigned.

The student has shown diligence, technical aptitude, and consistent progress throughout the internship period. The work reported herein is the original work carried out by the student and is a true record of the internship experience.

Internal Supervisor:

Ms. Surbhi Chhabra

Assistant Professor CSE Dept.

Institute of Eng. and Tech

JK Lakshmipat University

Jaipur-302026

External Supervisor:

Mr. Sajal Jain

(The Correct Steps)

Ghaziabad, Uttar Pradesh 201019

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to everyone who supported me during my Practice School II internship.

I would like to thank Mr. Sajal Jain, Founder of Correct Steps Consultancy, for providing the opportunity to work on a practical software development project and for offering valuable guidance during the internship.

I also extend my gratitude to my Internal Mentor **Ms. Surbhi Chhabra** from the Department of Computer Science & Engineering at JK Lakshmi pat University for their academic guidance and encouragement throughout the internship period.

Finally, I would like to thank my colleagues and peers who helped create a collaborative learning environment that enhanced my practical understanding of full-stack application development.

CONTENTS

TITLE

COVER

CERTIFICATION

ACKNOWLEDGEMENTS

ABSTRACT

LIST OF FIGURES

CHAPTER 1: ABOUT THE ORGANIZATION

1.1 Goals & Missions

1.2 Services & Products Offered

CHAPTER 2: INTRODUCTION

CHAPTER 3: TOOLS & TECHNOLOGY STACK

CHAPTER 4: ROLES AND RESPONSIBILITIES

CHAPTER 5: WORK DONE

CHAPTER 6: FUTURE PLAN

CHAPTER 7: CONCLUSION

REFERENCES

ABSTRACT

This report presents the mid-semester progress of the Practice School II internship carried out at **Correct Steps Consultancy**.

The primary focus of the internship was the development of an **Online Test Series Platform**, designed to provide a scalable and automated solution for conducting online examinations. The system supports multiple user roles including College, and Student, each with dedicated dashboards and controlled access.

The platform enables institutions to schedule tests, manage question papers, and analyze student performance through automated marking and result generation. Students can view upcoming tests, attempt examinations within specified time limits, and track their performance through detailed result reports and leaderboards.

The project focuses on implementing secure authentication, structured database design, and scalable full-stack architecture using modern web technologies. The system aims to improve transparency, efficiency, and accessibility in digital examination systems.

Keywords: Online Examination System, Test Series Platform, Automated Evaluation, Role-Based Dashboards, Web Application Development.

1.ABOUT THE ORGANIZATION

1.1 Goals & Missions

Correct Steps Consultancy is a modern and innovative company that helps businesses and individuals grow through a mix of creative, technical, and consulting services. The company was founded in 2022 by **Sajal Jain and Karan Gupta**. CSC is based in **Padova**, Italy, and works with clients from various industries to offer smart, custom-made solutions.

Mission

At Correct Steps Consultancy, the main goal is to guide clients step-by-step to success by using advanced technology, creative ideas, and strong business strategies.

1.2 Services & Products Offered

CSC offers a wide range of services in simple categories:

Business Consulting

- Market Research and Analysis
- Strategic Planning
- Financial Planning
- Client Acquisition
- Brand Development

Graphic Design

- Logo and Visual Design
- Portfolio Reviews
- Workshops and Tutorials
- Trend Insights and Feedback

Digital Marketing

- SEO (Search Engine Optimization)
- Social Media Management
- Content Creation
- Online Branding

CAD Design

- 2D and 3D Modeling
- Product and Architectural Design
- Drafting and Rendering
- Quality Checks

Engineering Design & Simulation

- Mathematical Modeling
- Design Validation
- Simulation using tools like ABAQUS and LS-DYNA

3D Printing Services

- Design to Print
- Material Selection
- High-Precision Printing
- Prototyping

2: INTRODUCTION

2.1 Internship Context

The Practice School II internship was undertaken at Correct Steps Consultancy during the academic year 2025-26. The internship provided an opportunity to apply theoretical knowledge gained in academic coursework to practical industry projects.

During the internship period, I was assigned a project involving the design and development of an Online Test Series Platform.

2.2 Project Overview – Online Test Series Platform

Traditional examination systems often rely on manual evaluation and lack transparency in result processing. To address these limitations, the Online Test Series Platform was designed as a scalable web-based system.

The platform allows organizations and educational institutions to manage examinations digitally through structured dashboards.

Project Overview – Ultraradx Web Application

As part of the internship, I also worked on developing a web application hosted at <https://ultraradx1.netlify.app/>. The project focused on building a responsive and user-friendly interface using modern frontend technologies. It demonstrates component-based UI development, responsive design, and efficient client-side navigation. The application was deployed on a cloud hosting platform to ensure accessibility and performance. Through this project, practical experience was gained in frontend development, deployment, and real-world web application workflows.

3: Tools & Technology Stack

3.1 Full Stack Development – Online Test Series Platform

Technology	Purpose
JavaScript	Core programming language used for frontend and backend development
React.js	Frontend library used to build interactive dashboards and user interfaces
Node.js	Runtime environment used for backend server development
Express.js	Backend framework used for building REST APIs and handling server-side routing
MongoDB	NoSQL database used to store user data, test information, questions, and results
Mongoose	Object Data Modeling (ODM) library used for managing MongoDB schemas and queries
VPS Server	Used for hosting the web application and backend services
JWT (JSON Web Tokens)	Used for secure authentication and managing user sessions
bcrypt	Used for password hashing and secure credential storage
Middleware	Used to protect backend routes and validate authentication tokens

4: ROLES AND RESPONSIBILITIES

4.1 Objectives

The primary objective of the internship was to design and develop a scalable web-based application for managing online examinations. The project aimed to implement a full stack solution that enables companies and educational institutions to conduct structured online tests with automated evaluation and performance tracking.

The key objectives of the project were:

- To design and develop a secure Online Test Series Platform.
 - To implement role-based dashboards for Company, College, and Student users.
 - To develop RESTful APIs for authentication, test management, and result processing.
 - To design efficient database schemas for storing user data, test details, and examination results.
 - To implement automated test evaluation and leaderboard generation.
 - To ensure secure user authentication and role-based access control.
-

4.2 Roles and Responsibilities Undertaken

During the internship, I was involved in multiple stages of the software development lifecycle, including system design, development, testing, and documentation.

Research and System Design:

Studied existing online examination platforms and analyzed their limitations to design a scalable and efficient system architecture.

Backend Development:

Developed backend APIs using Node.js and Express.js to manage user authentication, test scheduling, question handling, and result processing.

Frontend Development:

Implemented interactive dashboards using React.js for Company, College, and Student users to manage tests and view results.

Database Design:

Designed database models using MongoDB and Mongoose to store user data, test information, questions, and examination results.

Authentication and Security:

Implemented secure login and registration systems using JWT authentication and password encryption.

Testing and Debugging:

Tested system workflows, verified API responses, and ensured accurate result generation and leaderboard updates.

WORK DONE

5.1 System Architecture

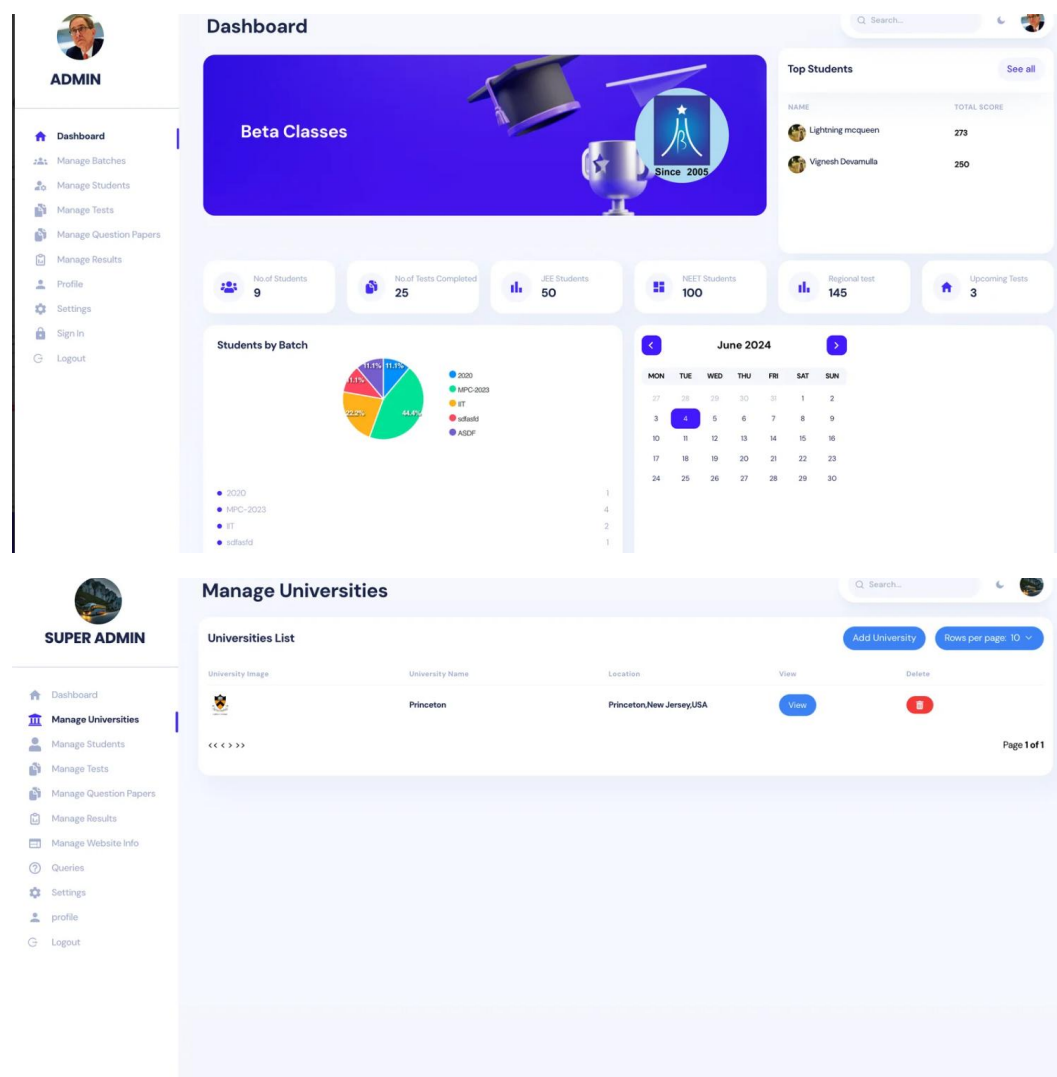
(Insert system architecture diagram here)

The system follows a three-layer architecture consisting of frontend, backend, and database layers.

5.2 Dashboard Development

The platform includes separate dashboards for Company, College, and Student users.

Each dashboard provides different features depending on user roles.



5.3 Test Management Module

The test management module allows colleges to create tests, upload question papers, and define test duration.



STUDENT

- Student Main Dashboard
- Upcoming Tests**
- Result
- Profile
- Settings
- Logout

Upcoming Tests



Beta Classes

Exam



Finals
 Score: 100
 Course: MPC
 Batch Name: MPC-2023

Start

Search...

Top Students [See all](#)

NAME	TOTAL SCORE
 Lightning mcqueen	273
 Vignesh Devamulla	250



ADMIN

- Dashboard
- Manage Batches
- Manage Students
- Manage Tests**
- Manage Question Papers
- Manage Results
- Profile
- Settings
- Sign In
- Logout

Manage Tests

Search...

[Add Test](#) Rows per page: 10

Test ID	Test Name	Test Stream	Max Marks	Question Paper Id	View	Delete
22	jee test week 2	MPC-2023	98	249	View	Delete
42	Mains third week	MPC-2023	100	44	View	Delete
20240226	Testing Test by K	MPC-2023	6	20240226	View	Delete
040	fouth week neet test	MPC-2023	99	444	View	Delete
050	mpc fourth week test	MPC-2023	100	444	View	Delete
90	sadfd	MPC-2023	100	90	View	Delete
1999	sdfdsf	MPC-2023	1	204	View	Delete
8990	fifth week	MPC-2023	1	204	View	Delete
9090	fifth week test	MPC-2023	2	912	View	Delete
1234567890	Land Mark Test	MPC-2023	2	7890	View	Delete

5.4 Automated Evaluation

The platform automatically evaluates student answers and generates results instantly after submission.

5.5 Leaderboard and Performance Tracking

Students can view their class ranking and performance through a dynamic leaderboard.

STUDENT

Student Main Dashboard

Upcoming Tests

Result

Profile

Settings

Logout

Result

Q Search...

Tests Results

Test Id	Exam Name	Marks	Max marks	Date	View
22	jee test week 2	3	98	2024-02-26	View
42	Mains third week	2	100	2024-02-25	View
20240226	Testing Test by K	3	6	2024-02-26	View
050	mpc fourth week test	4	100	2024-02-27	View
90	sadfd	1	100	2024-02-27	View
1999	sdfdsf	1	1	2024-02-28	View
8990	fifth week	1	1	2024-02-28	View
9090	fifth week test	12	2	2024-02-28	View
1234567890	Land Mark Test	3	2	2024-02-29	View
9087	9087	1	2	2024-02-29	View

STUDENT

Student Main Dashboard

Upcoming Tests

Result

Profile

Settings

Logout

Student Main Dashboard

Q Search...

Class Ranking60

Papers Solved21

Upcoming Tests4

Performance

Line chart showing performance from Jan to Dec for Math, Science, and English.

June 2024

Calendar for June 2024.

ADMIN

Dashboard

Manage Batches

Manage Students

Manage Tests

Manage Question Papers

Manage Results

Profile

Settings

Sign In

Logout

Manage Students

Q Search...

Students History

Add Student

Rows per page: 10

Students Image	Roll No	First Name	Last Name	Email	View	Delete
	1001	Lightning	mcqueen	mcqueen890@gmail.com	View	
	1002	Lightning	mcqueen	mcqueen89@gmail.com	View	
	21	shih	tzu	shih@gmail.com	View	
	12	Bhanu Prakash	Koppunoor	kbhanuprakashreddy143@gmail.com	View	
	33	sdf	fs	sdfdf@gmail.com	View	
	234	asdf	asdf	SDF@GMAIL.COM	View	
	5443	asdf	asdf	asdfsdf@gmail.com	View	
	89	Vignesh	Devamulla	vignesh@gmail.com	View	
	1003	Sanjay Kumar	Bhupathi	sanjay@gmail.com	View	

<< >>

Page 1 of 1

5.2 Week-Wise Log

Week Activities Completed

- Week 1** Focused on understanding the project requirements and defining the overall system architecture for the **Online Test Series Platform**. Studied existing online examination systems and identified limitations such as manual evaluation, lack of automation, and limited performance analytics. Designed the initial architecture including frontend, backend, and database components.
- Week 2** Set up the development environment and initialized the project repository using Git. Implemented the backend server using **Node.js and Express.js** and created initial API routes for user authentication and test management. Designed database schemas for users, tests, questions, and results using **MongoDB and Mongoose**.
- Week 3** Developed the frontend interface using **React.js**. Designed responsive dashboards for Company, College, and Student users. Implemented navigation using React Router and integrated frontend components with backend APIs for data retrieval and form submission.
- Week 4** Implemented **secure authentication mechanisms** including login and registration functionality using **JWT authentication and bcrypt password encryption**. Developed role-based access control to ensure that different users can access only their respective dashboards.
- Week 5** Implemented the **test management module**, allowing colleges to create tests, upload question papers, and define test durations. Developed features for students to view upcoming tests and attempt exams through a structured test interface.
- Week 6** Implemented the **automated evaluation system** that calculates scores after test submission. Developed the **leaderboard and performance tracking module** where students can view their rankings and results. Conducted testing of system workflows and improved API performance.
- Week 7** Worked on additional development tasks including building a web application hosted at <https://ultraradx1.netlify.app/>. The application demonstrates frontend development skills and integration with modern web technologies. Final testing and debugging of the Online Test Series Platform were performed along with documentation of system architecture and project implementation.

FUTURE PLAN

The future development of the Online Test Series Platform will focus on integrating advanced AI features to enhance examination monitoring, performance analysis, and user experience.

One of the key improvements will be the implementation of **AI-based monitoring or supervising an exam to prevent cheating.**, which can monitor student activity during examinations using webcam analysis and behavioral detection to prevent cheating. Another planned feature is **AI-driven performance analytics**, which will analyze student test results and provide personalized insights and recommendations for improvement.

Additionally, **AI-powered question generation and difficulty analysis** can be integrated to automatically generate test questions and classify them based on difficulty levels. These enhancements will improve the scalability, security, and intelligence of the platform while providing a more efficient and transparent examination environment.

CONCLUSION

This mid-semester report summarizes the progress made during the Practice School II internship at **Correct Steps Consultancy**. The primary focus of the internship was the design and development of an **Online Test Series Platform** that enables organizations and educational institutions to conduct structured online examinations efficiently. The system integrates role-based dashboards, secure authentication, automated test evaluation, and performance tracking to improve transparency and efficiency in the examination process.

Through this project, significant progress was made in implementing the core modules of the platform, including backend API development, frontend dashboard interfaces, database design, and secure authentication mechanisms. The development process provided practical experience in full stack application development using modern technologies and helped in understanding real-world software architecture and deployment practices.

Overall, the internship has provided valuable hands-on exposure to system design, full stack development, and project implementation. The remaining work will focus on enhancing system features, improving performance, and integrating advanced capabilities such as AI-based monitoring and analytics to further strengthen the platform.

References

[2] Tilkov, Stefan, and Steve Vinoski. 2010. *Node.js: Using JavaScript to Build High-Performance Network Programs*. *IEEE Internet Computing*, Vol.14, Issue 6, pp.80–83.

[4] MongoDB Inc. 2023. *MongoDB: The Developer Data Platform for Modern Applications*. MongoDB Technical Whitepaper.

[5] Hardt, Dick. 2012. *The OAuth 2.0 Authorization Framework*. Internet Engineering Task Force (IETF) RFC 6749.

[6] Jones, Michael B., Bradley Campbell, and Nat Sakimura. 2015. *JSON Web Token (JWT)*. Internet Engineering Task Force (IETF) RFC 7519.

[7] Sharma, R., and P. Singh. 2021. *Design and Implementation of Online Examination Systems Using Web Technologies*. *International Journal of Computer Applications*, Vol.174, Issue 15, pp.25–31.